Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

How is the interference pattern in Young's double slit experiment affected if the sodium (yellow) light is replaced by red light of the same intensity?

(A) The fringes will vanish

(B) The fringes will become brighter

(C) The fringe width will decrease

(D) The fringe width will increase

Q.2

Two coherent sources of intensity ratio $100:1$, interfere. What is the ratio of the intensity between the maxima and minima in the interference pattern?

(A) $10:1$

(B) $5:2$

(C) $121:81$

(D) $11:9$

Q.3

In a single slit diffraction experiment, the width of the slit is made double its original width. Then the central maximum of the diffraction pattern will become.

(A) narrower and fainter

(B) narrower and brighter

(C) broader and fainter

(D) broader and brighter.

Q.4

A parallel beam of monochromatic light is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of the incident beam. At the first minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of the slit is

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(A) zero

(B) $\frac{\pi}{2}$

(C) π

(D) 2π

Q.5

Young's double slit expetiment is performed using light of wavelength λ. One of the slits is covered by a thin glass sheet of refractive index μ at this wavelength. The smallest thickness of the sheet to bring the adjacent minimum to the centre of the screen is

(A) $\frac{\lambda}{2(\mu-1)}$

(B) $\frac{\lambda}{(\mu-1)}$

(C) $\frac{\lambda}{2\mu}$

(D) $\frac{\lambda}{\mu}$

Q.6

A single-slit diffraction pattern is obtained using a beam of red light. What happens if the red light is replaced by blue light?

(A) There is no change in the diffraction pattern.

(B) Diffraction fringes become narrower and crowded together.

(C) Diffraction fringes become broader and farther apart.

(D) The diffraction pattern disappears.

Q.7

What happens if the monochromatic light used in Young's double slit experiment is replaced by white light?

(A) All bright fringes become white

(B) All bright fringes have colours between violet and red

(C) Only the central fringe is white, all the other fringes are coloured

(D) No fringes will be observed.

Q.8

A parallel beam of fast moving electrons is incident normally on a narrow slit. A screen is placed at a large

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distance from the slit. If the speed of the electrons is increased, which of the following statements is correct?

(A) Diffraction pattern is not observed on the screen in the case of electrons

(B) The angular width of the central maximum of the diffraction pattern will increase

(C) The angular width of the central maximum will decrease

(D) The angular width of the central maximum will remain the same.

Q.9

Two waves of intensities I and $4I$ superpose, then the maximum and minimum intensities are

(A) $5I, 3I$

(B) $9I, I$

(C) $9I, 3I$

(D) $5I, I$

Q.10

Interference pattern is obtained with two coherent light sources of intensities I and $4I$. The intensity at a point where the phase difference is $\pi/2$ is

(A) I

(B) $2I$

(C) $3I$

(D) $5I$

Q.11

A ray of light is incident on glass at the polarising angle i_{p}. The angle between the reflected and refracted rays will be

(A) between 0° and 90°

(B) 90°

(C) between 90° and 180°

(D) $2i_{\mathrm{p}}$

Q.12

Which one of the following statements is correct? The refractive index of a given piece of glass is

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(A) less for violet than for red light

(B) more for blue than for green light

(C) less for green than for yellow light

(D) the same for all colours of light.

Q.13

In Young's double slit experiment, the intensity of the maxima is I. If the width of each slit is doubled the intensity of the maxima will be

(A) $\frac{I}{2}$

(B) I

(C) $2I$

(D) $4I$

Q.14

Which one of the following waves cannot be polarized?

(A) radio waves

(B) X-rays

(C) transverse waves in a string

(D) longitudinal waves in a gas.

Q.15

A beam of light consisting of two wavelengths $450\ \text{nm}$ and $750\ \text{nm}$ is used to obtain interference fringes in Young's double slit experiment. The separation between the slits is $1.0\ \text{mm}$ and the distance between the plane of the slits and the screen is $100\ \text{cm}$. The least distance from the central maximum where the bright fringes due to both the wavelengths coincide is

(A) $2.00\ \text{mm}$

(B) $2.25\ \text{mm}$

(C) $2.50\ \text{mm}$

(D) $2.75\ \text{mm}$

Q.16

What is the effect on the interference fringes in Young's double slit experiment if the source slit is moved closer to the double slit plane?

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(A) The fringe width increases

(B) The fringe width decreases

(C) The fringes become more distinct

(D) The fringes become less distinct.

Q.17

Two coherent light sources of intensity ratio n are employed in an interference experiment. The ratio of the intensities of the maxima and minima in the interference pattern is

(A) $\left(\frac{n+1}{n-1}\right)$

(B) $\left(\frac{n+1}{n-1}\right)^2$

(C) $\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$

(D) $\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)^2$

Q.18

When a ray of light is incident on a glass slab at an angle of 60°, the angle between the reflected and refracted rays is 90°. The refractive index of glass is

(A) 1.5

(B) $\sqrt{2}$

(C) $\sqrt{3}$

(D) 2.0

Q.19

Monochromatic light of wavelength $500\ \text{nm}$ is used in Young's double slit experiment. One of the slits is covered by a glass sheet of thickness $2.0\times 10^{-2}\ \text{mm}$ and refractive index 1.5. The number of fringes shifted by the introduction of the sheet is

(A) 14

(B) 16

(C) 18

(D) 20

Q.20

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White light is used to illuminate the two slits in Young's double slit experiment. The distance between the slits is b and the screen is at a distance $d\ (>> b)$ from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of the missing wavelengths are

(A) $\lambda = \frac{b^2}{2d}$

(B) $\lambda = \frac{2b^2}{d}$

(C) $\lambda = \frac{b^2}{3d}$

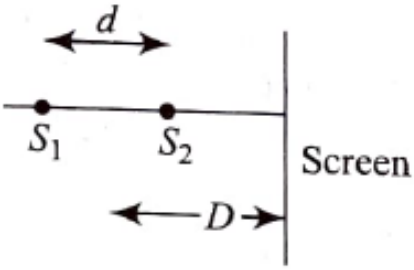
(D) $\lambda = \frac{2b^2}{3d}$

Q.21

The angle of polarization for any medium is 60°, what will be critical angle for this?
If critical angle C is represented as $\sin C = \frac{M}{\sqrt{3}}$, then M=?

Q.22

Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 300 makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and Screen intensities of the two beams are 1 I_B respectively, then I_A/I_B equals (the result is expressed as 1/ x. Then x is ?)



Q.23

The angular width $\Delta\theta$ of the central maximum in the diffraction pattern of a slit illuminated by light of wavelength $500\ \text{nm}$ is measured. The angular width of the central maximum is found to increase by 25% when the same slit is illuminated by another light of wavelength λ'. Wavelength λ' (in nm) is

Q.24

Unpolarized light of intensity 32Wm^{-2} passes througy three polarizers such that transmission axes of the first and second polarizer makes an angle 30° with each other and the transmission axis of the last polarizer is crossed with that the first. The intensity of final emerging light will be

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Q.25

A beam of light of wavelength $600\ \text{nm}$ from a distant source falls on a single slit $1.0\ \text{mm}$ wide and the resulting diffraction pattern is observed on a screen $2\ \text{m}$ away. The distance between the first dark fringes on either side of the central bright fringe is (Integral value of the result has to be taken)

Q.26

A parallel beam of monochromatic light of wavelength $600\ \text{nm}$ falls normally on a slit of width $0.5\ \text{mm}$. The resulting diffraction pattern is observed on a screen placed $50\ \text{cm}$ from the slit. The linear separation between the first minimum and the first maximum on the same side of the central maximum is (the result is represented as 0.x mm , then x is.?)

Q.27

In Young's double slit experiment, the fringe width is $2.0\ \text{mm}$, The separation between the 9 th bright fringe and the second dark fringe from the centre of the fringe system will be

Q.28

A plane wavefront $\left(\lambda = 6 \times 10^{-7}\ \text{m}\right)$ falls on a slit $0.4\ \text{mm}$ wide. A convex lens of focal length $0.8\ \text{m}$ placed behind the slit focusses the light on a screen. What is the linear diameter of second maximum?

Q.29

In an interference experiment, 20 th order maximum is observed at a point on the screen when light of wavelength $480\ \text{nm}$ is used. If this light is replaced by light of wavelength $600\ \text{nm}$, the order of the maximum at the same point will be

Q.30

In Young's double slit experiment the distance d between the slits S_1 and S_2 is $1.0\ \text{mm}$. What should the width of each slit be so as to obtain 10 maxima of the two slit interference pattern within the central maximum of the single slit diffraction pattern? (the result is represented as 0.x mm, then x is.?)

Answers & Solutions

Q.1 Answer:

The fringe width will increase

Solution:

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The wavelength of red light is greater than that of yellow light. Since $\beta = \lambda D/d$, the fringe width β will increase. Hence the correct choice is (d).

Q.2 Answer:

$121 : 81$

Solution:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}$$
$$= 100 + 1 + 2\sqrt{100 \times 1} = 121$$

Intensity at minima is

$$I' = I_1 + I_2 - 2\sqrt{I_1 I_2} = 81$$
$$\therefore \quad \frac{I}{I'} = \frac{121}{81} = 1.49$$

Hence the closest choice is (c).

Q.3 Answer:

narrower and brighter

Solution:

The angular width of the central maximum is $2\lambda/a$ where a is the width of the slit. If the value of a is doubled, the angular width of the central maximum decreases to half its earlier value. This implies that the central maximum becomes much sharper. Furthermore if a is doubled, the intensity of the central maximum becomes two times. Thus the central maximum becomes much sharper and brighter.

Q.4 Answer:

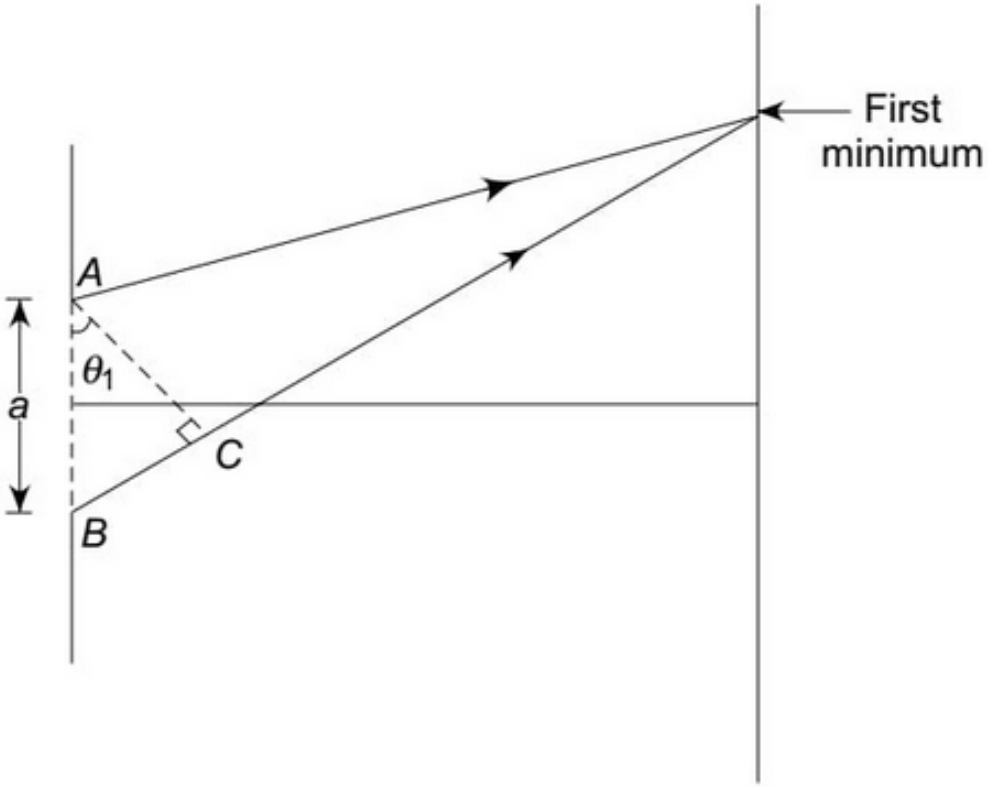
2π

Solution:

It follows from Fig. that the path difference at the first minimum between rays coming from A and B is

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$\Delta = BC = a \sin\theta_1$



But $\sin\theta_1 = \lambda/a$. Therefore, $\Delta = a \times \lambda/a = \lambda$. A path difference of λ corresponds to a phase difference of 2π, which is choice (d).

Q.5 Answer:

$\frac{\lambda}{2(\mu-1)}$

Solution:

If t is the thickness of the glass sheet, the fringes are displaced by an amount given by

$\Delta y = (\mu - 1)\frac{tD}{d}$

In order to bring the adjacent minimum to the centre of the screen (i.e. to bring the first dark fringe the central bright fringe), the fringes must be displaced by half the fringe width, i.e.

$\Delta y = \frac{\beta}{2} = \frac{1}{2}\frac{\lambda D}{d}$

Hence $(\mu - 1)\frac{tD}{d} = \frac{1}{2}\frac{\lambda D}{d}$

or $t = \frac{\lambda}{2(\mu-1)}$**, which is choice (a).**

Q.6 Answer:

Diffraction fringes become narrower and crowded together.

Solution:

The wavelength of blue light is less than that of the red light. Hence the angular width of the maxima will decrease which means that the fringes become narrower and crowded together.

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Q.7 Answer:

Only the central fringe is white, all the other fringes are coloured

Solution:

White light consists of colours between violet and red. The wavelength λ is the shortest for violet 1 ight and the longest for red light. At the central fringe, the path difference for all colours is zero. Therefore, at the centre of the screen all colours superpose to give a white fringe. Hence the correct choice is (c).

Q.8 Answer:

The angular width of the central maximum will decrease

Solution:

De Broglie wavelength of electron is

$\lambda = \frac{h}{p} = \frac{h}{mv}$

If speed v of electron is increased, momentum $p(=mv)$ will increase. Hence wavelength λ will decrease. Now, the angular width of the central maximum of the diffraction pattern is 2θ where θ is given by

$\sin\theta = \frac{\lambda}{a}$

where a is the width of the slit. Thus, if λ decreases, θ and hence 2θ will decrease. Therefore, the correct statement is (c).

Q.9 Answer:

$9I, I$

Solution:

Given $I_1 = I$ and $I_2 = 4I$. Now

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$
$$= I + 4I + 2\sqrt{4I^2} = 9I$$
$$\text{and} \quad I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2} = 5I - 4I = I$$

Hence the correct choice is (b).

Q.10 Answer:

$5I$

Solution:

The correct choice is (d). Use

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$
$$\text{Put} \quad I_1 = I, I_2 = 4I \quad \text{and} \quad \phi = \frac{\pi}{2}.$$

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Q.11 Answer:

90°

Solution:

The correct choice is (b).

Q.12 Answer:

more for blue than for green light

Solution:

The refractive index of glass decreases with increase in wavelength. In VIBGYOR, the wavelength of violet light is the shortest and that of the red light is the longest. Hence the correct choice is (b).

Q.13 Answer:

$2I$

Solution:

$I = I_{\text{max}} = I_1 + I_2 + 2\sqrt{I_1 I_2}$

When the width of each slit is doubled, I_1 becomes $2I_1$ and I_2 becomes $2I_2$. Therefore,

$$I' = I'_{\text{max}} = 2I_1 + 2I_2 + 2\sqrt{2I_1 \times 2I_2}$$
$$= 2\left(I_1 + I_2 + 2\sqrt{I_1 I_2}\right) = 2I_{\text{max}} = 2I$$

Hence the correct choice is (c).

Q.14 Answer:

longitudinal waves in a gas.

Solution:

Only transverse waves can be polarized. Radiowaves, X-rays and waves on strings are transverse and hence they can be polarized. Longitudinal waves such as sound waves or waves in springs cannot be polarized. Hence the correct choice is (d).

Q.15 Answer:

2.25 mm

Solution:

Let nth bright fringe of wavelength $\lambda_1 = 450$ nm coincide with the m th bright of $\lambda_2 = 750$ nm. Then

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$$y = \frac{n\lambda_1 D}{d} = \frac{m\lambda_2 D}{d} \quad \ldots(\mathrm{i})$$

$$\Rightarrow n\lambda_1 = m\lambda_2$$

$$\Rightarrow \frac{n}{m} = \frac{\lambda_2}{\lambda_1} = \frac{750\ \mathrm{nm}}{450\ \mathrm{nm}} = \frac{5}{3} \quad \ldots(\mathrm{ii})$$

The smallest integral values of n and m which satisfy (ii) are $n = 5$ and $m = 3$. From (i) we have

$$y_{\min} = \frac{n\lambda_1 D}{d}$$

$$= \frac{5 \times \left(450 \times 10^{-9}\right) \times 1.0}{\left(1.0 \times 10^{-3}\right)}$$

$$= 2.25 \times 10^{-3}\ \mathrm{m} = 2.25\ \mathrm{mm}$$

Q.16 Answer:

The fringes become less distinct.

Solution:

Let x be the width of the source slit and X the distance between the source slit and the plane of the two slits. For interference fringes to be distinctly visible, the condition $x/X < \lambda/d$ should be satisfied. If x is too large (i.e. the source slit is too wide) or if X is too small (X is the distance between the source slit and the two slits) the requirement $x/X < \lambda/d$ may be violated and fringes will no longer be distinct. The reason is that the interference patterns due to various parts of the source slit overlap. Consequently, the minima will not be totally dark and fringe pattern becomes indistinct. However, as long as the fringe pattern remains visible, a change in x or X has no effect on the fringe width β

Q.17 Answer:

$\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)^2$

Solution:

Given $\frac{1_1}{I_2} = n$. Therefore, the amplitude ratio is

$\frac{A_1}{A_2} = \sqrt{n}$

Now $I_{\max} = (A_1 + A_2)^2$ **and** $I_{\min} = (A_1 - A_2)^2$

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$$\therefore \quad \frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{\left(\frac{A_1}{A_2} + 1\right)^2}{\left(\frac{A_1}{A_2} - 1\right)^2}$$

$$= \frac{(\sqrt{n} + 1)^2}{(\sqrt{n} - 1)^2}$$

Hence the correct choice is (d).

Q.18 Answer:

$\sqrt{3}$

Solution:

Angle of incidence $= i$**, angle of refraction** $r = 90 - i$ **Hence the refractive index is**

$\mu = \frac{\sin i}{\sin r} = \frac{\sin i}{\sin(90^\circ - i)} = \frac{\sin i}{\cos i} = \tan i$

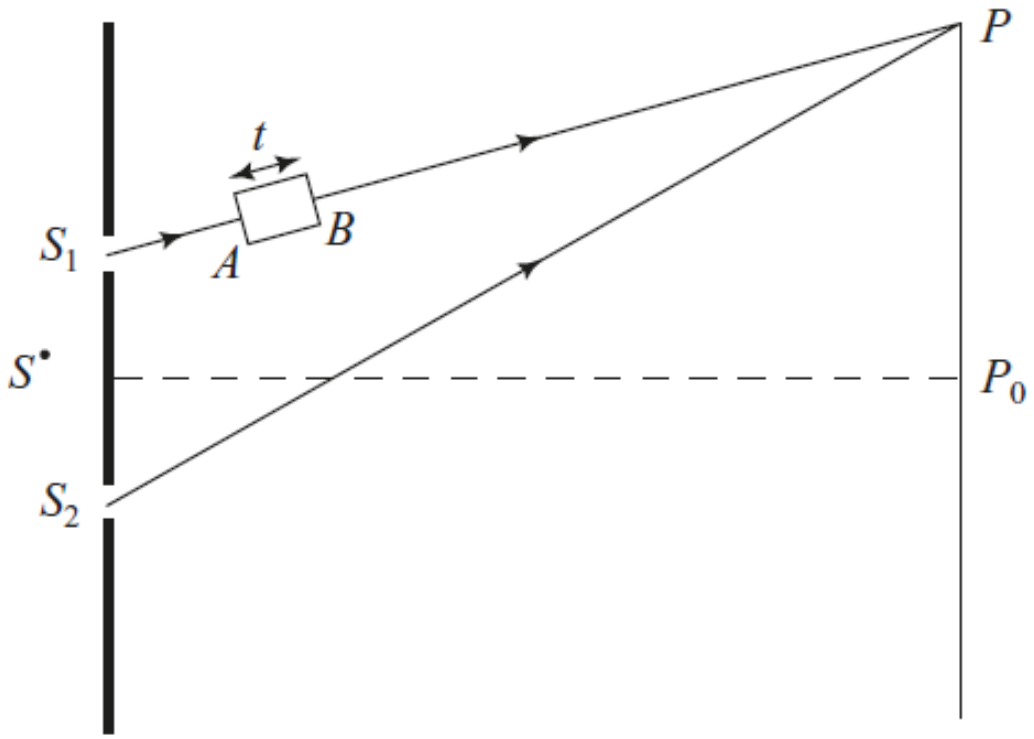
$\therefore \quad \mu = \tan 60^\circ = \sqrt{3}$**, which is choice (c).**

Q.19 Answer:

20

Solution:

When light traverses a sheet of thickness t **and refractive index** μ**, the optical path travelled** $= \mu t$



Path difference in the absence of sheet is

$\Delta_1 = S_2P - S_1P$

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Path difference when the sheet is introduced is

$$\begin{aligned}\Delta_2 &= S_2P - [S_1A + \mu(AB) + BP] \\ &= S_2P - [S_1A + BP + AB - AB + \mu(AB)] \\ &= S_2P - [S_1P + (\mu - 1)AB] \\ &= S_2P - S_1P - (\mu - 1)t\end{aligned}$$

$\therefore$ Change in optical path is

$$\Delta = \Delta_1 - \Delta_2 = (\mu - 1)t$$

If the optical path changes by one wavelength, one fringe will shift. Therefore, the number of fringes shifted due to introduction of sheet is

$$\frac{(\mu-1)t}{\lambda} = \frac{(1.5-1)\times\left(2.0\times10^{-5}\ \mathrm{m}\right)}{(500\times10^{-9}\ \mathrm{m})} = 20$$

Q.20 Answer:

$\lambda = \frac{b^2}{3d}$

Solution:

The missing wavelengths are

$$\begin{aligned}\lambda &= \frac{b^2}{(2m+1)d}; m = 0, 1, 2, 3, \cdots \text{ etc.} \\ &= \frac{b^2}{d}, \frac{b^2}{3d}, \frac{b^2}{5d}, \cdots \text{ etc.}\end{aligned}$$

Hence the correct choice is (c).

Q.21 Answer:

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Solution:

By using $\mu = \tan\theta_\pi$ mplies $\mu = \tan 60 = \sqrt{3}$

also $C = \sin^{-1}\left(\frac{1}{\mu}\right) \Rightarrow C = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Q.22 Answer:

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Solution:

$$\mathrm{I_A}\cos^2 30 = \mathrm{I_B}\cos^2 60$$

$$\frac{\mathrm{I_A}}{\mathrm{I_B}} = \frac{1}{3}$$

Q.23 Answer:

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Solution:

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Given

$$\Delta\theta' = \Delta\theta + 25\% \text{ of } \Delta\theta$$
$$= \Delta\theta + \frac{25}{100}\Delta\theta = 1.25\Delta\theta$$
$$\therefore \quad \frac{\lambda'}{\lambda} = \frac{\Delta\theta'}{\Delta\theta} = 1.25$$
$$\Rightarrow \quad \lambda' = 1.25\lambda = 1.25 \times 500 \text{ nm} = 625 \text{ nm}$$

Q.24 Answer:

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Solution:

If θ is the angle between the transission axes of first polaroid P_1 and second P_2 while ϕ between the transmission axes of second polaroid P_2 and third P_3, then according to given problem. $\theta + \phi = 90^\circ$ or $\phi = (90^\circ - \theta)$.....(1)\nNow if I_0 is the intensity of unpolarized light incident on polaroid P_1, the intensity of light transmitted through it,\n$I_1 = \frac{1}{2}I_0 = \frac{1}{2}(32) = 16\frac{W}{m^2}$\nNow as angle between transmission axes of polariods P_1 and P_2 is θ, in a accordance with Malus law, intensity of light transmitted through P_2 will be\n$I_2 = I_1\cos^2\theta = 16\cos^2\theta \ldots\ldots(3)$\nAnd as angle between transmission axes of P_2 and P_3 is ϕ, light transmitted through P_3 will be\n$I_3 = I_2\cos^2\phi = 16\cos^2\theta\cos^2\phi \ldots..(4)$\nAccording to given problem, $I_3 = 3W/m^2$\nSo, $4(\sin 2\theta)^2 = 3 i.e., \sin 2\theta = (\sqrt{3}/2)$ or\n$(2\theta = 60^\circ, i.\text{ e. }, \theta = 30^\circ$

Q.25 Answer:

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Solution:

The angular separation between the m th dark fringe and the central bright fringe is given by

$$a\sin\theta_n = n\lambda$$

For the first dark fringe, $n = 1$. Therefore

$$a\sin\theta_1 = \lambda$$

or $\sin\theta_1 = \lambda/a$. Since $\lambda << a, \sin\theta_1 \simeq \theta_1$. Hence $\theta_1 = \lambda/a$. This is also the angular separation between the central bright fringe and the first dark fringe on the other side of the central bright fringe. Hence, the angular separation between the first dark fringes on either side of the central bright fringe $= 2\theta_1 = 2\lambda/a$ Therefore, their separation at distance $d = 2$ m is

$$\frac{2\lambda d}{a} = \frac{2 \times \left(600 \times 10^{-9} \text{ m}\right) \times 2 \text{ m}}{1.0 \times 10^{-3} \text{ m}}$$
$$= 2.4 \times 10^{-3} \text{ m} = 2.4 \text{ mm}$$

Q.26 Answer:

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Solution:

For first minÄ±mum, $(a =$ **width of the slit** $)$

$a \sin\theta_1 = \lambda$

$\Rightarrow \quad \sin\theta_1 = \frac{\lambda}{a} \quad \Rightarrow \quad \theta_1 = \frac{\lambda}{a} \quad (\because \quad \lambda << a)$

For first maximum,

$a \sin\theta_2 = \frac{3\lambda}{2} \Rightarrow \theta_2 = \frac{3\lambda}{2a}$

Angular separation between them is

$\Delta\theta = \theta_2 - \theta_1 = \frac{3\lambda}{2a} - \frac{\lambda}{a} = \frac{\lambda}{2a}$

∴ Linear separation is

$$\Delta x = D\Delta\theta = \frac{D\lambda}{2a}$$

$$= \frac{0.5 \times \left(600 \times 10^{-9}\right)}{2 \times \left(0.5 \times 10^{-3}\right)} = 0.3 \text{ mm}$$

Q.27 Answer:

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Solution:

The distance of the n**th bright fringe from the central fringe is**

$y_n = n\lambda\frac{D}{d} = n\beta$

where $\beta = \frac{\lambda D}{d}$ **is the fringe width.**

$\therefore \quad y_n = 9\beta$ **...(1)**

The distance of the m **th dark fringe from the central fringe is**

$y_n' = \left(m - \frac{1}{2}\right)\frac{\lambda D}{d} = \left(m - \frac{1}{2}\right)\beta$

$y_2' = \frac{3}{2}\beta \quad \ldots(2)$

From Eqs. (1) and (2) we get

$$y_9 - y_2' = 9\beta - \frac{3}{2}\beta = \frac{15}{2}\beta$$

$$= \frac{15}{2} \times 2.0 \text{ mm} = 15 \text{ mm}$$

Q.28 Answer:

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Solution:

For secondary maxima $\mathrm{d}\sin\theta = \frac{5\lambda}{2}$

$\Rightarrow \mathrm{d}\theta = \mathrm{d} \cdot \frac{\mathrm{x}}{\mathrm{D}(\approx \mathrm{f})} = \frac{5\lambda}{2}$

$\Rightarrow 2\mathrm{x} = \frac{5\lambda\mathrm{f}}{\mathrm{d}} = \frac{5 \times 0.8 \times 10^{-7}}{4 \times 10^{-4}} = 6 \times 10^{-3} \text{ m} = 6 \text{ mm}$

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Q.29 Answer:

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Solution:

The position of the nth order maximum is given by

$y_n = \frac{n\lambda D}{d}$

For a given point y_n is fixed. Since D and d are also fixed, $n\lambda =$ constant, i.e. $n_1\lambda_1 = n_2\lambda_2$.

Hence $n_2 = n_1 \frac{\lambda_1}{\lambda_2} = \frac{20\times 480}{600} = 16$, which is choice (a).

Q.30 Answer:

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Solution:

Let the width of each slit be a. The linear separation between m bright fringes in the double slit experiment is

$y_m = \frac{n\lambda D}{d}$

Since $y << D$, the angular separation between m bright fringes will be

$\theta_m = \frac{y_n}{D} = \frac{n\lambda}{d}$

For 10 bright fringes we have

$\theta_{10} = \frac{10\lambda}{d}$ **...(i)**

Now the angular width of the principal maximum in the diffraction pattern due to a slit of width a is

$2\theta_1 = \frac{2\lambda}{a}$ **...(ii)**

Equating (i) and (ii), we get

$\frac{10\lambda}{d} = \frac{2\lambda}{a}$

or $a = \frac{d}{5} = \frac{1.0\ \text{mm}}{5} = 0.2\ \text{mm}$

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